# BASIC 4

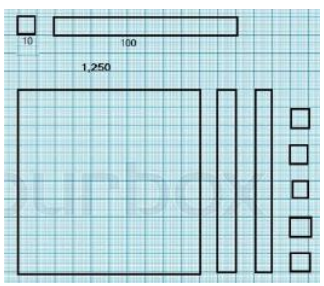
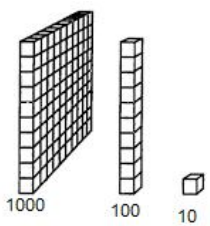
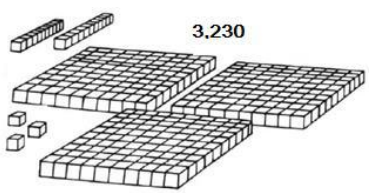


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**Basic 4**

**Strand 1: NUMBER**

**Sub-strand 1: Counting, Representation & Cardinality**

| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.1.1**<br>**Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000.** | **B4.1.1.1.1 Model number quantities, place value for multi-digit using graph sheets or multi-base materials up to 100,000**<br>E.g.1. Ask learners to model number quantities up to 100,000 using graph sheets or multi-base materials. For instance, with multi-base block , a cube = 100 unit; a rod = 1000; a flat = 10,000 and a block = 100,000; learners model 32,300 with the appropriate materials<br>E.g.2 Ask learners to model the number 12,500 shading graph sheet square as shown below, where 2cm×2cm square represents 10 units.<br>  <br>  <br>  | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Cultural Identity to Precision. |

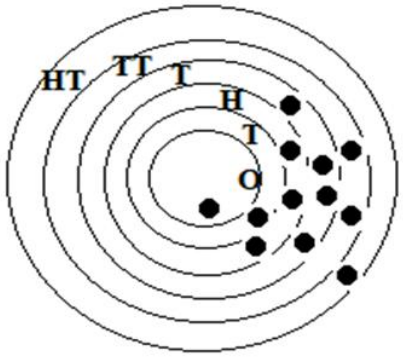


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| **B4.1.1.1**<br>**Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000. CONT'D** | E.g. 3 Give learners teacher-made token currency notes [¢1, ¢10 & ¢100 notes] on different coloured-paper and ask them to model or pick given amounts up to ¢10,000; e.g. ¢2,480<br>  <br><br>**B4.1.1.1.2 Read and write numbers in figures and in words up 100,000**<br>E.g. 1 Play the place value number wheel game: Use the hundred thousand number wheel to generate 6-digit numbers and represent the number generated on a place value frame. Throw a number of pebbles (or stones) onto the number wheel, identify the number of pebbles landing in each place value ring, and state the number generated in the wheel (i.e. 2 landed in the ten-thousand' ring making the number twenty-thousand or 20,000).<br>NB: The representations to use also include verbal, and numerals. Learners must match number word cards to the figures | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Cultural Identity |



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| **B4.1.1.1**<br>**Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000**<br>**CONT'D** |  <br><br>Place value number wheel<br><br>Place value chart<br><br>Hundred-thousands: 1<br>Ten-thousands: 2<br>Thousand: 4<br>Hundreds: 3<br>Tens: 1<br>Ones: 1<br><br>E.g. 2 Ask learners to write given numbers on the expanded form of the number:<br>14031=10,000 + 4,000 +30 + 1<br><br>E.g. 3 Display a number chart or number line with multiples of 500 between 10000 and 50000 and lead learners to identify numbers in different positions around a given number. | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Look for Patterns |



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<td>B4.1.1.1<br>Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000.<br>CONT'D</td>
<td>
B4.1.1.1.3 Identify numbers in different positions around a given number in a number chart<br><br>
E.g. 1 Display a number chart in multiples of 500 between 10,000 and 100,000 and lead learners identify numbers in different positions around a given number. Put learners in pairs and give each group a number grid and have them identify numbers in different positions around a chosen number.
<table>
<tr><td>10,000</td><td>10,500</td><td>11,000</td><td>11,500</td><td>12,000</td><td>12,500</td></tr>
<tr><td>20,000</td><td>20,500</td><td>21,000</td><td>21,500</td><td>22,000</td><td>22,500</td></tr>
<tr><td>30,000</td><td>30,500</td><td>31,000</td><td>31,500</td><td>32,000</td><td>32,500</td></tr>
<tr><td>40,000</td><td>40,500</td><td>41,000</td><td>41,500</td><td>42,000</td><td>42,500</td></tr>
<tr><td>50,000</td><td>50,500</td><td>51,000</td><td>51,500</td><td>52,000</td><td>52,500</td></tr>
<tr><td>60,000</td><td>60,500</td><td>61,000</td><td>61,500</td><td>62,000</td><td>62,500</td></tr>
</table>
B4.1.1.1.4 Compare and order whole numbers up to 10,000 and represent comparisons using the symbols "<", "=", ">"<br><br>
E.g. 1 Learners identify numbers which are 1,000 and 10,000 more or less than a given five-digit or six-digit number. E.g. 2400 is 1,000 less than 3400. Learners use words such as: "equal to" "greater than" "less than" and later use symbols such as "<", "=", ">" to compare numbers up to 100,000 taking into consideration the place value of each digit in the given number. E.g. $1200 = 1200;\ 27345 > 26355,$<br><br>
E.g. 2 Learners work together in their groups to order a given set of numbers in ascending or descending order verbally and in writing. E.g. $1020, 1025, 2673, 2873,$<br><br>
E.g. 3 Give two numbers between 10,000 and 100,000 to each group and encourage learners to say as many things as possible about the two numbers. For instance, 21345 and 21534; 21345 is less than (smaller than) than 21534 or 21534 is bigger than (greater than) 21345, or 21345 is almost 200 less than 21534 etc.
</td>
<td>Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Look for Patterns</td>
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<tr><td>B4.1.1.1<br>Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000<br>CONT'D</td><td>
B4. 1.1.1.5 Round (off, up, down) whole numbers up to 10000 to the nearest, thousands, hundreds and tens<br><br>
E.g. 1 Learners round off numbers to the nearest 1000, 100 and 10. For instance; 9500 is approximately 10,000 and 9100 as approximately 9,000<br><br>
E.g. 2 Learners round up and round down to estimate sums and differences<br><br>
NB: To round down learners consider the smaller of the two approximated numbers, while round up considers the bigger of the two numbers. The table below may bring the meaning of these two concepts clearer<br><br>
<table>
<tr><td>14765</td><td>Round up</td><td>Round down</td><td>Round off</td></tr>
<tr><td>to the nearest ten</td><td>14970</td><td>14760</td><td>14970</td></tr>
<tr><td>to the nearest hundred</td><td>14800</td><td>14700</td><td>14800</td></tr>
<tr><td>to the nearest thousand</td><td>15000</td><td>14000</td><td>15000</td></tr>
</table>
<br>
E.g. 3 Learners work together to round up and round down to estimate sums and differences. For instance. $230+160$ as approximately 400 because $230$ is approximately $200$ while $160$ is approximately $200$, therefore $200+200=400$
</td><td>Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Look for Patterns</td></tr>
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| **B4.1.1.1**<br>**Demonstrate an understanding of quantities and place value for multi-digit whole numerals up to 100,000. CONT'D** | **B4.1.1.1.6. Skip count forwards and backwards in 50s and 100s up to and from 10000**<br><br>E.g. 1 Put learners into convenient groups. A learner mentions a number and another makes skip count in 50s or 100s to include the fifth count. For instance.<br>Learner 1: Shout out "240" …. Learner 2: 290, 340, 390, 440, 490, etc.<br>Learner 3; Shout out "1285" *skip counting down in 100s*<br>Learner 4; 1185, 1085, 985, 885, 785….etc.<br>E.g. 1 Skip count forwards and backwards by 50s and between 1000 and 10000 that are multiples of 50s and 100s but make an error or leave out a number. Challenge learners to identify or correct error | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Look for Patterns |
| **B4.1.1.2.**<br>**Demonstrate understanding of Roman numerals up XXX (30)** | **B4.1.1.2.1 Develop an understanding of Roman Numeral system up XXX (i.e. 30)**<br><br>E.g. 1 Display Roman numeral charts (1-30) arranged in sequential order and lead learners to identify the numerals. Learners identify the main characters of the roman numerals used to build the table up to 30 i.e. I, II, III, IV, V, X,<br><br>Call out a numeral and have learners point at it from the chart<br><br>E.g. 2 Have learners match the Roman numerals to the Hindu-Arabic numerals for instance I = 1; V= 5: X= 10, XV= 15. Mention some numerals randomly and have learners point at it on the chart | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision;<br>Look for Patterns and Relationships |

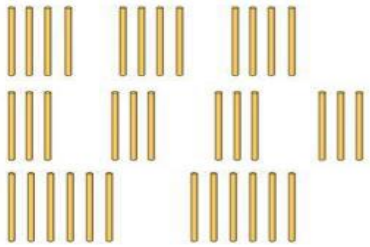


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| **B4.1.1.2. Demonstrate understanding of Roman numerals up XXX (30) CONT'D** | **B4.1.1.2.2 Count and convert Hindu Arabic numerals to Roman numerals up to 30 and vice versa**<br>E.g. 1 Learners read the chart sequentially forwards and backwards, vertically (up and down), zig-zag, diagonally and randomly without pointing to the numbers. Invite 2-3 learners to read to the class. Call a Roman numeral and ask learners to write<br>E.g. 2 Give learners a numeral in the Hindu Arabic system and have learners convert to roman numeration and vice versa for instance XXIV = 24<br>**I V X L C**<br>**1 5 10 50 100**<br>1 = I 10 = X<br>2 = II 20 = XX<br>3 = III 30 = XXX<br>4 = IV 40 = XL<br>5 = V 50 = L<br>6 = VI 60 = LX<br>7 = VII 70 = LXX<br>8 = VIII 80 = LXXX<br>9 = IX 90 = XC<br>10 = X 100 = C<br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

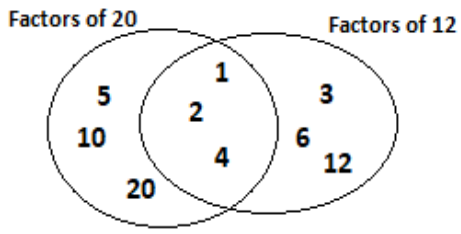


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| **B4.1.1.3 Demonstrate an understanding of factors, multiples and squared numbers** | **B4.1.1.3.1 Determine set of factors of a given numbers up to 50**<br><br>E.g. 1 Learners make formation of sets of equal object from a given quantity. For example with 12 straws Learners make groupings such as:<br><br>  <br>$3\ groups\ of\ 4\ \Rightarrow 3,4$<br>$4\ groups\ of\ 3 \Rightarrow 4,3$<br>$2\ groups\ of\ 6 \Rightarrow 2,6$<br><br>E.g. 2 Learners pair grouping and the number of equal objects for each formation as factors;<br><br>4 groups of 3 straws (4, 3)<br>6 groups of 2 straws (6, 2)<br>1 group of 12 straws (1, 12)<br>3 groups of 4 straws (3, 4)<br>2 groups of 6 straws (2, 6)<br>12 groups of 1 straw (12, 1)<br><br>E.g. 3 Learners collect from the pairs and form a set as the factors of the given number. For example the set of factors of $12 = \{1, 2, 3, 4, 6, 12\}$ | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships, Cultural Identity |

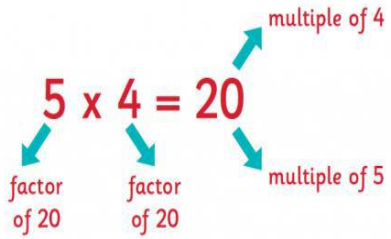
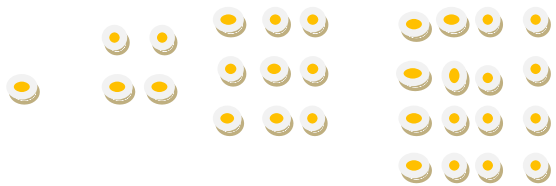


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| **B4.1.1.3 Demonstrate an understanding of factors, multiples and squared numbers CONT'D** | **B4.1.1.3.2 Determine the highest common factor (HCF) of any two whole numbers between 1 and 50.**<br><br>E.g. 1 Learners list the factors of two or more given whole numbers; 12 and 24 to list the factors. That is: $12 = \{1, 2, 3, 4, 6, 12\} and\ 24 = \{1, 2, 3, 4, 6, 8, 12, 24\}$<br><br>Learners determine the set of the common factors E.g. The common set factors of 12 and 24 = $\{1,2,3,4,6,12\}$. Learners select the highest common factor of 12 and 24 as 12<br><br>E.g. 2 Learners may use Venn diagrams to find the common factors and then the highest common factor by placing the factors in the regions of the shapes. The highest common factor of 12 and 20, in the diagram is 4<br><br>  <br><br><br>Note: Do not introduce formal set theory and notation. Numbers in the common regions of the two shapes.<br><br>**B4. 1.1.3.3. Determine the lowest common multiple (LCM) of at least any two given numbers up to 100.**<br><br>E.g. 1 Learners identify and the list the set multiples of given numbers using skip counting E.g. The set of multiples 5 = { 5, 10 15……..100} and the etc. of multiples of 10 = {10, 20, 30 , ……..100}<br><br>E.g. 2 Learners list the set of common multiples of any two or more given numbers between 0 and 100. E.g. The set of common multiples of 5 and 10 ={10, 20, 30, … 100}. Learners select the least of the common multiple of the given numbers, and identify the least common multiple (LCM). The LCM of 5 and 10 is 10. | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision |

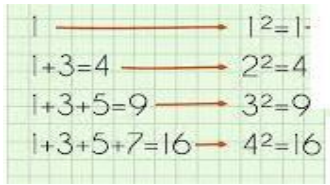
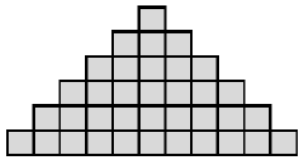
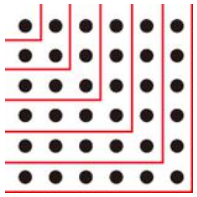
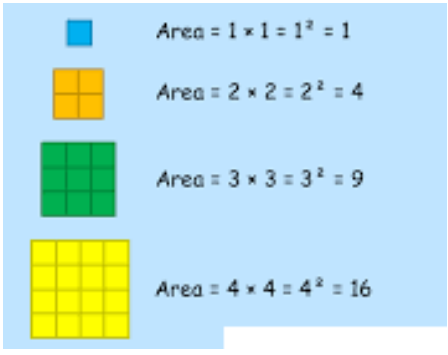


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| **B4.1.1.3 Demonstrate an understanding of factors, multiples and squared numbers CONT'D** | **B4.1.1.3.4 Recognise the relationship between factors and multiples.**<br>E.g. 1 Factors represent the numbers that we can multiply to get a bigger number called the product or multiple. Since $12 = 4 \times 3$, we can say 4 and 3 are factors of 12; and 12 is a multiple of 3 or a multiple of 4. Break given multiples into factors and pairs of factors of given numbers.<br>E.g. 2 Investigate even and odd numbers.<br>  <br>E.g. 3 Investigate numbers that are multiples of 2, 3, 4 and 5. How do you know a number is a multiple of 2? a multiple of 3? a multiple of 4? and a multiple of 5? (This is also known as the divisibility test).<br>**B4.1.1.3.5 Generate and analyse patterns in square numbers**<br>E.g. 1 Put learners into convenient groups and provide each group adequate number of bottle tops, an A4 sheet and a 30cm ruler and ask them to generate square numbers by continuing the arrangement shown below up to the 10th number; [i.e. 1, 4, 9, 16, 25, 36 … )<br>  <br>1 4 9 16 …… | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |



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| **B4.1.1.3 Demonstrate an understanding of factors, multiples and squared numbers CONT'D** | E.g. 2 Investigate the relationship between square numbers and the sum of consecutive odd numbers.<br>  <br><br>  <br>  <br>**B4.1.1.3.6 Represent square numbers using factors**<br>E.g. 1 Learners draw a square whose side is 1 unit by 1 unit and determine the area; repeat the activity for a 2 unit, 3 unit and 4 unit squares.<br>E.g. 2 Learners work in groups to continue the pattern to include the factors of 144.<br>(Key Concept: when a number has been multiplied by itself we say the answer is a square number hence $1\times \mathbf{1}=\mathbf{1}; \mathbf{2}\times\mathbf{2}=\mathbf{4};\ \mathbf{3}\times\mathbf{3}=\mathbf{9}$ and we can write three squared as $\mathbf{3}\times\mathbf{3}=\mathbf{3^2}$)<br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships: Cultural Identity |

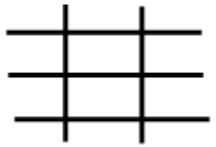


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| **B4.1.1.4**<br>**Interpret negative and positive numbers in context** | **B4.1.1.4.1 Describe real life situations using positive and negative values**<br>E.g. 1 Brainstorm learners on happenings which may be represented with positive and negative numbers (e.g. having savings with a bank and owing a bank; profit and loss etc.)<br>Movement on the number line to the right and left of zero<br>**B4.1.1.4.2 Count forwards and backwards with positive and negative whole numbers through zero**<br>E.g. 1 Learners count from a given interval of positive and negative numbers on the number line. For example, count from 10 backward through zero to negative -8. | Learners develop:<br><br>Critical Thinking;<br>Justification of Ideas;<br>Attention to Precision |



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| **B4.1.2.1**<br><br>**Recall basic multiplication fact up to $12 \times 12$** | **B B4.1.2.1.1 Determine basic multiplication facts up to $12 \times 12$**<br><br>E.g. 1. To find $2 \times 3 = ?$ Learners arrange 2 straws vertically (legs) and put across it with 3 horizontal ones (arms). Learners count the number of intersections<br><br>  <br>$2 \times 3 = 6$<br><br>E.g. 2. Learners model different multiplication sentences, for example $3 \times 4$. Learners mention the number of dots (intersections) for other learners to model fence<br><br>**Note:** Alternatively, learners may draw vertical lines to represent the legs and horizontal lines across them to represent the arms and then count the number of intersections as the product<br><br>E.g. 3. Draw a 6 by 6 multiplication chart for the numbers 5 to 10 and use it as a game board to play the 3-in-a-line game with a pair of dice or playing cards. Players take turns in throwing a pair of dice and mark (or cover) the product made in a throw with his/her marker (or counter). The winner is the one who obtains three products in a line.<br><br>   <br><br>| | 5 | 6 | 7 | 8 | 9 | 10 |<br>|---|---|---|---|---|---|---|<br>| 5 | 10 | 15 | 20 | 25 | 30 | 35 |<br>| 6 | 12 | 18 | 24 | 30 | 36 | 42 |<br>| 7 | 14 | 21 | 28 | 35 | 42 | 49 |<br>| 8 | 16 | 24 | 32 | 40 | 48 | 56 |<br>| 9 | 18 | 27 | 36 | 45 | 54 | 63 |<br>| 10 | 20 | 30 | 40 | 50 | 60 | 70 |<br><br>E.g. 4. Learners find doubles of given numbers; find squares of given numbers; and skip count in 4, 5, and 8 | Learners develop:<br><br>Problem solving skills; Critical Thinking; Justification of Ideas; Collaborative learning; Attention to Precision; Look for Patterns and Relationships: Global Citizenship |

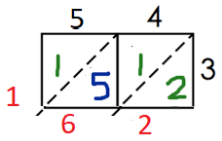
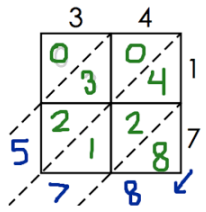


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| **B4.1.2.2**<br>**Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 81 and related division facts** | **B4.1.2.2.1 Apply mental mathematics strategies and number properties, such as skip counting from a known fact using doubling or halving using patterns in the 9s facts using repeated doubling or halving to determine answers for basic multiplication facts to 81 and related division facts**<br>E.g. 1 Describe the mental mathematics strategies used to determine a given basic fact, such as<br>– skip count up by one or two groups from a known fact (e.g., if 5 × 7 = 35, then 6 × 7 is equal to 35 + 7 and 7 × 7 is equal to 35 + 7 + 7)<br>– skip count down by one or two groups from a known fact (e.g., if 8 × 8 = 64, then 7 × 8 is equal to 64 – 8 and 6 × 8 is equal to 64 – 8 – 8)<br>– doubling (e.g., for 8 × 3 think 4 × 3 = 12, and 8 × 3 = 12 + 12)<br>– patterns when multiplying by 9 (e.g., for 9 × 6, think 10 × 6 = 60, and 60 – 6 = 54; for 7 × 9, think 7 × 10 = 70, and 70 – 7 = 63)<br>– repeated doubling (e.g., if 2 × 6 is equal to 12, then 4 × 6 is equal to 24 and 8 × 6 is equal to 48)<br>– repeated halving (e.g., for 60 ÷ 4, think 60 ÷ 2 = 30 and 30 ÷ 2 = 15)<br>– relating division to multiplication (e.g., for 64 ÷ 8, think 8 × … = 64)<br>E.g. 2 Recall multiplication facts to 81 and related division facts | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Attention to Precision; Look for Patterns and Relationships |



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<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B4.1.2.2<br>Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 81 and related division facts CONT'D</td><td>B4.1.2.2.2 Apply mental mathematics strategies for multiplication, such as annexing then adding zero halving and doubling using the distributive property<br>E.g. 1 Determine the products when one factor is a multiple of 10, 100, or 1000 by annexing zero or adding zeros (e.g., for 3×200 think of 3×2 and then add two zeros)<br>E.g. 2 Apply halving and doubling when determining a given product (e.g., 32×5 is the same as 16×10)<br>E.g. 3 Apply the distributive property to determine a given product involving multiplying factors that are close to multiples of 10 (e.g., 29×7 = (30×7) – (1×7) = 203</td><td>Learners develop:<br><br>Problem solving skills; Critical Thinking; Justification of Ideas; Attention to Precision; Look for Patterns and Relationships</td></tr>
<tr><td>B4.1.2.3 Demonstrate an understanding of multiplication (2 or 3-digit by 1-digit )</td><td>B4. 1.2.3.1 Multiply multi-digit numbers efficiently<br>E.g. 1. Multiplication of whole numbers using the "expand and box" method (partial decomposition method). Lead learners to multiply a 3-digit number by a 1-digit number as shown below, for example $448 \times 2 = ?$<br><table><tr><td>×</td><td>400</td><td>40</td><td>8</td></tr><tr><td>2</td><td>800</td><td>80</td><td>16</td></tr></table>$448 \times 2 = (400 \times 2) + (40 \times 2) + (8 \times 2)$<br>$= 800 + 80 + 16 = 800 + 80 + (10 + 6)$<br>$= 800 + 90 + 6 = 896$<br>  <br><br>E.g. 2. Multiplication of whole numbers using the lattice method. Have learners draw a 2 by 1 rectangular box for solving 54 × 3.<br>E.g. 3. Have learners draw a 2 by 2 rectangular box for solving 34 × 17.<br>  <br></td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships</td></tr>
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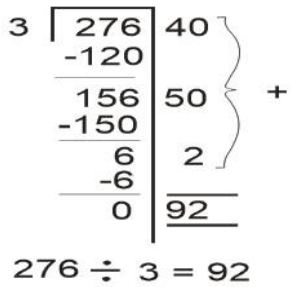
| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.2.4**<br>**Recall basic division fact up to 100** | **B4. 1.2.4.1 Determine basic division fact up to 81**<br>E.g. 1. Investigate numbers that are multiples of 2, 3, 4 and 5. How do you know a number is divisible by 2, 3, 4 and 5 (This is also known as the divisibility test).<br>E.g. 2. Draw a 6 by 6 multiplication chart and use it as a game board to play the 3-in-a-line game with a pair of dice or playing cards. Players take turns in throwing a one dice and mark (or cover) a number that can be divided by the results (i.e. a quotient). The winner is the one who obtains three numbers in a line.<br>   | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Global Citizenship |
| **B4.1.2.5**<br>**Demonstrate an understanding of division (2- or 3-digit by one digit number)** | **B4.1.2.5.1 Divide 2-digit numbers by 1-digit number efficiently**<br>E.g. 1. Division as repeated subtraction (using the long division method). Explain division as a way of repeatedly subtracting a divisor number from a given dividend until there is none left and then determining the number of times the divisor is taken from the dividend. For example, $25 \div 5 = ?$ Learners count the number of times 5 was subtracted from 25, which is 5 times. Hence, $\mathbf{25 \div 5 = 5}$ | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning |

| | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|
| 5 | 10 | 15 | 20 | 25 | 30 | 35 |
| 6 | 12 | 18 | 24 | 30 | 36 | 42 |
| 7 | 14 | 21 | 28 | 35 | 42 | 49 |
| 8 | 16 | 24 | 32 | 40 | 48 | 56 |
| 9 | 18 | 27 | 36 | 45 | 54 | 63 |
| 10 | 20 | 30 | 40 | 50 | 60 | 70 |

25
−5 1
20
−15 1
15
−5 1
10
−5 1
5
−5 1
00 5

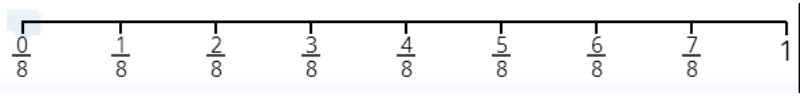
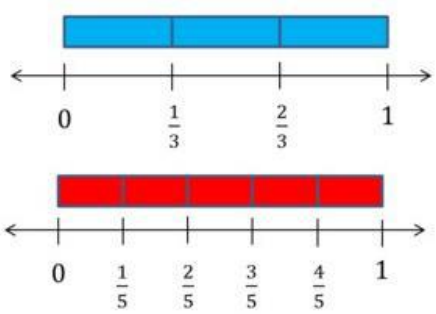


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| | E.g. 2. Division using the estimation of multiples of 10, 100 and others of the divisor. For Example to solve 276 ÷ 3 = ?<br><br>Learners use estimations: (they may have to use multiples of divisor to select a convenient estimate) "About how many groups of 3 can fit into 276?" The estimation used here is 40. So 3 × 40 = 120.<br><br>Subtracting 120 from 276 to get 156. The next estimation used is 50. So 3 × 50 = 150. Subtracting 150 from 156 to get 6. The next estimation used is 2. And 2 × 3 = 6. Subtracting 6 from 6 to get 0. There is nothing more to share. To find the final answer add the estimations: $40 + 50 + 2 = 92$. Therefore $276 \div 3 = 92$<br><br>  <br><br><br>Note: This strategy is nicknamed the "Big 7" because the division box looks like a 7. | |
| **B4.1.2.6**<br><br>**Translate and solve word problems involving the four basic operations on whole numbers** | **B4.1.2.6.1 Solve multi-step word problems involving the four basic operations**<br><br>E.g. 1. Learners model mathematical statements from a given word problem involving addition and multiplication and solve using the strategies learnt<br><br>E.g. 2. Learners model mathematical statements from a given word problem involving division and subtraction and solve using the strategies learnt<br><br>E.g. 3. Learners role play a given word problem involving addition and multiplication and solve<br><br>E.g. 4. Learners role play a given word problem involving subtraction and division and solve<br><br>E.g. 5. Learners model mathematical statements from a given word problem involving division and multiplication and solve using the strategies learnt The relationship between operations and the use of calculator and spreadsheet to assess the reasonableness of answers should be stressed | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; |

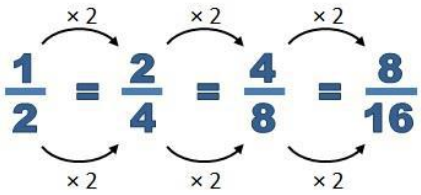
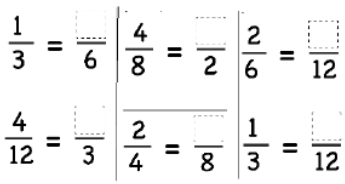
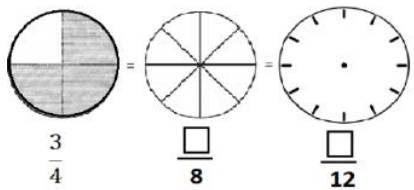


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.3.1**<br><br>**Develop an understanding of equivalent and improper fractions**<br>~~12 × 12~~ | **B4.1.3.1.1 Generate unit fractions and locate a unit fraction, e.g. one-eighth, on a number line by defining the interval from 0 to 1 as the whole and partitioning it into 8 equal parts and that each part has size $\frac{1}{8}$.**<br><br>E.g. 1. Use number line to locate one eight by defining the interval from 0 to 1 as the whole and partitioning it into 8 equal parts<br><br>  <br><br><br>E.g. 2. Ask pupils to draw several lines, 30 units each, and mark the ends 0 and 1. For each line ask them to partition the interval from 0 to 1 into each of the following unit fractions $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{6}$, and $\frac{1}{10}$.<br><br>E.g. 3. Ask pupils to use the number chart they have drawn to read fractions that are equivalent e.g. $\frac{1}{3} = \frac{2}{6} = \frac{3}{6}$. …<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision |

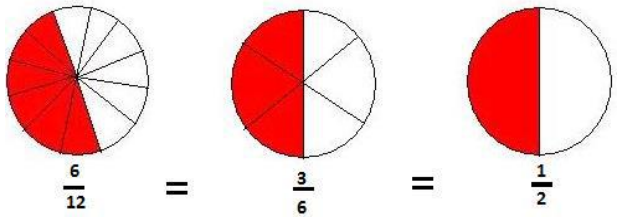


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.3.1**<br>**Demonstrate an understanding of strategies for comparing, adding and subtracting fractions (same denominator, or one being a multiple of the others)**<br>~~12 × 12~~ | **B4.1.3.1.2 Recognise and name equivalent fractions using pictorial representations and number line to determine the Lowest Common Denominator (LCD).**<br><br>E.g. 1. Use fraction chart to illustrate the concept of equivalent fractions (i.e. equal fractions that have different fractional units or denominators) i.e. one-half is 2-fourths, 3-sixth or 4-eighths.<br><br>E.g. 2. Illustrate the relationship between fractions and their equivalences by observing the different fractional units or denominators (i.e. the numerator is multiplied by the number of units the denominator is further broken into)<br><br>  <br><br><br>  <br><br><br>E.g. 3. Ask pupils to colour fractions that are equivalent to given fractions or write more equivalent fractions for give n fractions<br><br>  <br><br><br>  <br><br><br>E.g. 4. To determine the equivalent fractions of given fractions, find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions that are comparable | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

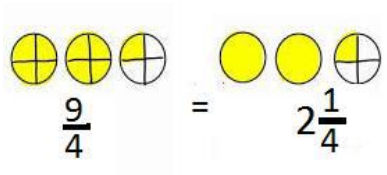
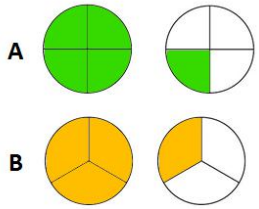
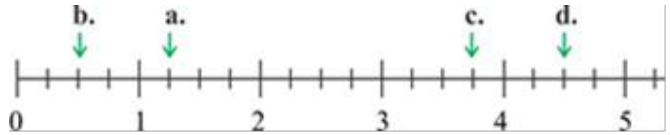


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|---|---|---|
| **B4.1.3.1 Demonstrate an understanding of strategies for comparing, adding and subtracting fractions (same denominator, or one being a multiple of the others) CONT'D** ~~12 × 12~~ | **B4.1.3.1.3 Find the simplest form of given fractions by dividing through by the highest common factor (i.e. by cancelling through by factors)**<br>E.g. 1. Use pictorial representations to illustrate the simplest form of a fraction; e.g. $\frac{6}{12} = \frac{3}{6} = \frac{1}{2}$<br>  <br><br>E.g. 2. Find the simplest form of a given fraction, divide through by the highest common factor.<br>$\frac{8}{12} = \frac{2}{3}$ (÷ 4)<br>or divide by 2, divide by 2, divide by 2, divide by 2: $\frac{8}{12} = \frac{2}{3}$<br>E.g. 3. Find the simplest form of a given fraction, express the numerator and denominators as factors and cancel through common factors.<br>$\frac{8}{12} = \frac{\cancel{2} \times \cancel{2} \times 2}{\cancel{2} \times \cancel{2} \times 3} = \frac{2}{3}$<br>E.g. 2. Find the simplest form of a given fraction, divide through by the highest common factor.<br>$\frac{8}{12} = \frac{2}{3}$ (÷ 4 ÷ 4)<br>or divide by 2, divide by 2, divide by 2, divide by 2: $\frac{8}{12} = \frac{2}{3}$ | Learners develop:<br><br>Problem solving skills; Critical Thinking; Justification of Ideas; Collaborative learning; Attention to Precision; Look for Patterns and Relationships |

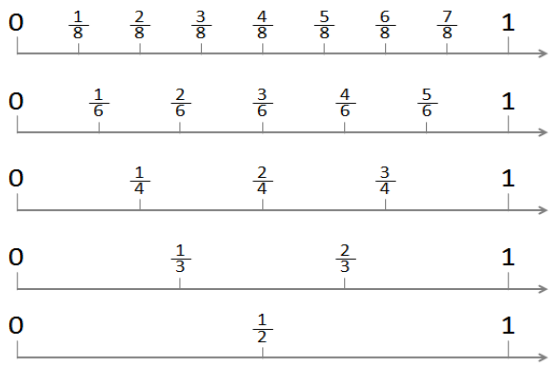


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.3.1 Demonstrate an understanding of strategies for comparing, adding and subtracting fractions (same denominator, or one being a multiple of the others) CONT'D** | E.g. 3. To find the simplest form of a given fraction, express the numerator and denominators as factors and divide through by common factors.<br>$\frac{8}{12} = \frac{\cancel{2} \times \cancel{2} \times 2}{\cancel{2} \times \cancel{2} \times 3} = \frac{2}{3}$<br>**B4.1.3.1.4 Recognise fractions that are greater than one (i.e. improper fractions), draw and label such fractions with their symbols**<br>E.g. 1. Use several pictorial representations (or card cut outs) to introduce improper fractions (i.e. where the fraction is bigger than 1 or the numerator in larger than the denominator). Examples are $\frac{9}{4}$ and $\frac{7}{6}$ (i.e. nine-fourths and seven-sixths) and show how they are expressed as mixed fractions (i.e. $2\frac{1}{4}$ and $1\frac{1}{6}$). Ask pupils to change given improper fractions to mixed fractions<br>  <br><br>E.g. 2. Ask pupils to express given coloured mixed fractions as improper fractions or draw and colour given improper fraction (e.g. seven-fourths, eleven-eights, etc.)<br>  <br><br>E.g. 3. Use number line to present more mixed fractions for pupils to identify and state their improper fractions<br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

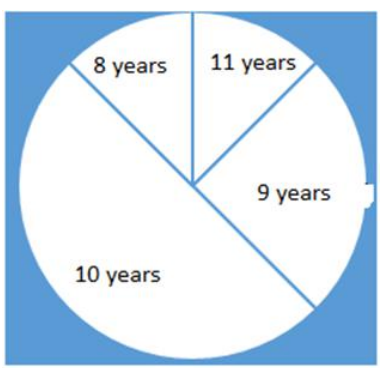


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| **B4.1.3.2**<br><br>**Demonstrate an understanding of strategies for comparing, adding and subtracting fractions (same denominator, or one being a multiple of the others)** | **B4.1.3.2.1 Compare and order fractions with like denominators by using pictorial representations and finding equivalent fractions using the Lowest Common Denominator (LCD)**<br><br>E.g. 1. Use number line or fraction chart to illustrate the concept of comparable fractional units or denominators; i.e. halves, fourths and eighths are comparable; i.e. half can be changed to equivalent fractions in fourths and eighths, and fourths can changed to eighths. Also thirds can changed to sixths.<br><br>E.g. 2. Use number line or fraction charts to compare and order unit fractions. Which is larger, $\frac{5}{8}$ and $\frac{3}{4}$ or $\frac{3}{6}$ and $\frac{4}{8}$? Arrange from smallest to largest, $\frac{3}{4}$, $\frac{2}{3}$ and $\frac{5}{6}$<br><br>  <br><br><br>E.g. 3. To arrange the fractions, $\frac{3}{4}$, $\frac{2}{3}$ and $\frac{5}{6}$, from largest to smallest, find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions<br><br>LCD for $\frac{3}{4}$, $\frac{2}{3}$ and $\frac{5}{6}$ is 12;<br>hence $\frac{3}{4} = \frac{9}{12}$; $\frac{2}{3} = \frac{8}{12}$; $\frac{5}{6} = \frac{10}{12}$ $\therefore$ the order is $\frac{5}{6}, \frac{3}{4}, \frac{2}{3}$ | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision |

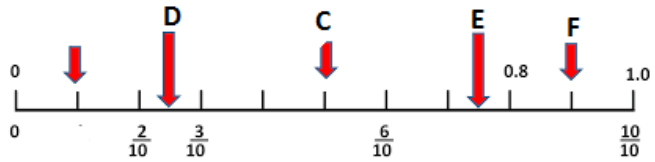


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | CORE COMPETENCIES |
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| **B4.1.3.2**<br><br>**Demonstrate an understanding of strategies for comparing, adding and subtracting fractions (same denominator, or one being a multiple of the others). CONT'D** | **B4.1.3.2.3 Provide examples of where fractions are used**<br><br>E.g. 1. Discuss the contexts below in which fractions are used in real life and provide materials for pupils to act them out<br>– Selling liquids – oil, kerosene, drinks, etc.<br>– Sharing pizza or birthday cake - half, quarter, eighths, etc.<br>– Buying objects in packets – egg, drinks, (half crate),<br>– Buying cloth for sewing dresses – half, one-fourths or quarter, etc.<br><br>  <br><br>E.g. 2. Tell the time by half/quarter past, and half/quarter to. Draw clock-faces with times (i.e. half/quarter past or to) for pupils to read the time<br><br>E.g. 3. Drawing circle graphs that represent halves, fourths and eighths. The graph shows the ages of pupils in Primary 4. If there are 40 pupils in the class, ask questions for pupils to interpret the graph<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills;<br>Critical Thinking;<br>Justification of Ideas;<br>Collaborative Learning;<br>Attention to Precision |



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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B4.1.4.1**<br>**Develop an understanding of decimals (tenths and hundredths) using concrete and pictorial representations and write decimal fractions as base ten numbers with place value.** | **B4.1.4.1.1 Describe and represent decimals (tenths and hundredths) concretely, pictorially, and symbolically**<br>E.g. 1. Use pictorial representations or number line to revise tenths and hundredths and ask pupils to identify the fractions (i.e. A, B, C, D, & E).<br>  <br><br>E.g. 2. Use the tenth and hundredth place value chart to illustrate how to write the fraction as a base ten number with the introduction of the decimal point "·" after ones place in the place value chart.<br>E.g. 3. Give pupils several common fractions (including improper fractions) to convert into tenths and hundredths and write their decimal names.<br>E.g. $7\frac{2}{5} = \frac{\square}{10} = \square$. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision |

**Tenth and Hundredth Place Value Chart**

| | Fraction | Is | · | $\frac{1}{10}s$ | $\frac{1}{100}s$ |
|---|---|---|---|---|---|
| **A** | $\frac{1}{4} = \frac{25}{100}$ | 0 | · | 1 | |
| **B** | $\frac{1}{8} = \frac{125}{1000}$ | 0 | · | 2 | 0 |
| **C** | $2\frac{1}{2} =$ | 0 | · | 5 | |
| **D** | $3\frac{3}{5} =$ | 0 | · | | |



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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B4.1.4.1**<br>**Develop an understanding of decimals (tenths and hundredths) using concrete and pictorial representations and write decimal fractions as base ten numbers with place value.** | **B4.1.4.1.2 Round decimals to the nearest tenth**<br>E.g. 1. Explain the rule for of rounding decimals, which is similar to that of rounding whole numbers. That is, rounding to the nearest tenth means that the rounded figure has one digit after the decimal mark. Rounding to the nearest hundredth means that the rounded figure has two digits after the decimal mark<br>E.g. 2. Use a table with several decimals fractions and explain the rule for of rounding decimals. Give pupils a table of decimal fractions to round to the nearest tenths or hundredths<br>E.g. 3. Ask students to change fractions to decimal writing their results to the nearest tenths or hundredths | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

**Rounding**

**85674.875**

**rounded to**

**10** ▶ **85670**

**100** ▶ **85700**

**1 decimal or $\frac{1}{10}$ place** ▶ **85674.9**

| | Fraction | Round to Nearest | Result |
|---|---|---|---|
| **A** | 0.38 | $\frac{1}{10}$ | 0.4 |
| **B** | 4.085 | $\frac{1}{10}$ | |
| **C** | 56.584 | $\frac{1}{10}$ | |
| **D** | 18.096 | $\frac{1}{100}$ | |
| **E** | 30.084 | $\frac{1}{100}$ | |



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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.4.1**<br>**Develop an understanding of decimals (tenths and hundredths) using concrete and pictorial representations and write decimal fractions as base ten numbers with place value** | **B4.1.4.1.3 Use models to explain the result of addition and subtraction of decimals (up to hundredths)**<br>E.g. 1. To add 0.64 and 0.39, means sum $\frac{64}{100} + \frac{39}{100}$ which is $\frac{64+39}{100} = \frac{103}{100}$ = 1.03.<br>That is, since both are hundredths fractions (i.e. have 2 decimal places) we can line up the decimal points to align the place values and add as whole numbers<br>E.g. 2. To add 0.6 and 0.39, one number is in tenths and other is in hundredths (i.e. one has 1 decimal place and the other 2 decimal places) it is important to line up the decimal points to align the place values and add as whole numbers<br>E.g. 3. To subtract 0.6 from 1.39, means $\frac{139}{100} - \frac{6}{10} = \frac{139}{100} - \frac{60}{100} = \frac{139-60}{100} = \frac{79}{100}$ = 0.79. That is, since one number is in tenths and other is in hundredths (i.e. one has 1 decimal place and the other 2 decimal places) it is important to line up the decimal points to align the place values and subtract as whole numbers. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

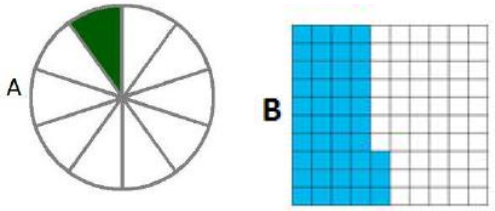
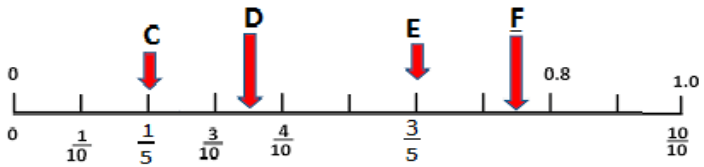
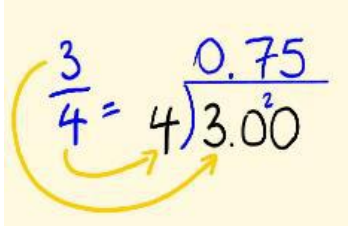
$$\begin{array}{r} 0.64 \\ +\ 0.39 \\ \hline 1.03 \end{array}$$

$$\begin{array}{r} 0.6 \\ +\ 0.39 \\ \hline 0.99 \end{array}$$

$$\begin{array}{r} 1.39 \\ -\ 0.6 \\ \hline 0.79 \end{array}$$

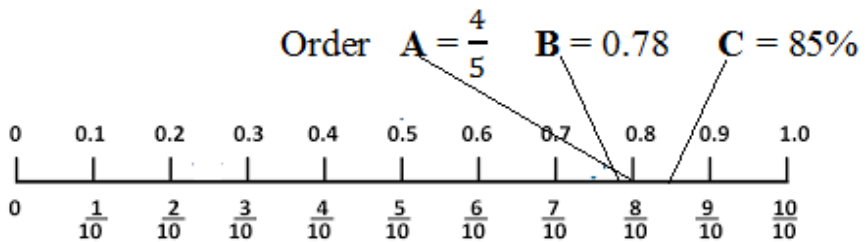


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.5.1**<br><br>**Demonstrate an understanding of percent (limited to whole numbers) concretely, pictorially, and symbolically)** | **B4.1.5.1.1 Model or recognise percent (as a fraction related to hundredths) using concrete models, pictorial representations and number line.**<br><br>E.g. 1. Use several pictorial representations or number line to introduce tenths and hundredths and ask pupils to identify the fractions (i.e. A, B, C, D, & E).<br><br>E.g. 2. Use long division method to convert and write fractions as decimals in the number chart.<br><br>  <br><br>  <br><br>E.g. 3. Give pupils several common fractions (including improper fractions) to convert into hundredths and write their decimal names or use long division; e.g. $\frac{3}{4} = \frac{75}{100}$ = 75%.<br><br>  <br><br>\| | Fraction | 2 Decimal Places | Percent \|<br>\| A \| $\frac{1}{10}$ \| 0.10 \| 10% \|<br>\| B \| $\frac{43}{100}$ \| 0.43 \| 43% \|<br>\| C \| \| 0.50 \| \|<br>\| D \| \| 0.35 \| \|<br>\| E \| \| \| \|<br>\| F \| ⋮ \| \| \| | Learners develop:<br><br>Problem solving skills; Critical Thinking; Justification of Ideas; Collaborative learning; Attention to Precision; Look for Patterns and Relationships |



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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.1.5.1**<br><br>**Demonstrate an understanding of percent (limited to whole numbers) concretely, pictorially, and symbolically) COND** | **B4.1.5.1.2 Compare and order a mixture of common, decimal and percent fractions (up to hundredths)**<br>E.g. 1. To compare and order a mixture of common, decimal and percent fractions, express them in one form (i.e. either common, decimal or percent); e.g. to order $\frac{4}{5}$, 0.78 and 85% (i) we can express all as decimals $\rightarrow \frac{4}{5} = \frac{80}{100}$; $0.78 = \frac{78}{100}$ and $85\% = \frac{85}{100}$, hence the order from least to the largest is 0.78, $\frac{4}{5}$ and 85%; (ii) we can also express all as percentages $\rightarrow \frac{4}{5} = \frac{80}{100}$ = 80%, $0.78 = \frac{79}{100}$ = 79%, and $= \frac{85}{100}$ = 85%, hence the order from least to the largest is 0.78, $\frac{4}{5}$ and 85%.<br><br>E.g. 2. To compare and order a mixture of common, decimal and percent fractions you can locate the fractions on the number and order them.<br><br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision; Look for Patterns and Relationships |

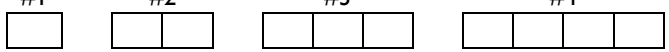


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<tr>
<td>B4.2.1.1<br><br>Demonstrate an understanding of how to identify and describe patterns found in tables and charts, including a multiplication chart</td>
<td>
B4.2.1.1.1 Describe the pattern found in a given table or chart<br><br>
E.g. 1. Using a hundred chart, ask students to colour each number as they skip count by 2s and describe the pattern created. Repeat for 3, 4, 5, 6, 7, 8, and 9. Ask students to describe what changes they notice as the numbers increase.<br>
Note to what extent students:<br>
(i) identify all, some or none of the multiples of a given number.<br>
(ii) are able to predict and extend the pattern.<br>
(iii) are able to describe the pattern by relating it to similar designs in the real world (e.g., students may say it looks like a checker board)<br><br>
B4.2.1.1.2 Determine the missing element(s) in a given table or chart<br><br>
E.g. 1. Provide students with a chart with missing numbers and ask them to identify the missing numbers and explain their reasoning<br><br>
B4.2.1.1.3 Identify the error(s) in a given table or chart<br><br>
E.g. 1. Provide a mapping table containing errors. Ask students to identify and explain where the pattern has errors
<table>
<tr><th>Number of Tables</th><th>Number of Chairs</th></tr>
<tr><td>1</td><td>4</td></tr>
<tr><td>2</td><td>8</td></tr>
<tr><td>3</td><td>12</td></tr>
<tr><td>4</td><td>18</td></tr>
<tr><td>5</td><td>20</td></tr>
<tr><td>6</td><td>24</td></tr>
</table>
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Pattern and Relationship</td>
</tr>
</table>



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<tr>
<td>B4.2.1.2.<br><br>Translate among different representations of a pattern, such as a table, a chart or concrete material</td>
<td>
B4.2.1.2.1 Create a concrete representation of a given pattern displayed in a table or chart<br>
E.g. 1. Provide a table showing a pattern (involving one operation) such as the one below. Students can complete the table and create a concrete representation using linking cubes or other manipulatives
<table>
<tr><td>Figure</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Number of Cubes</td><td>3</td><td>6</td><td>9</td><td>?</td><td>?</td><td>?</td><td>?</td><td>?</td></tr>
</table>
B4.2.1.2.2 Create a table or chart from a given concrete representation of a pattern.<br>
E.g. Present learners with a geometric design series, such as the one shown below. Ask them to extend the pattern and record it in a T-chart. Ask learners what the 10th step would be. What would the 12th step be? What would the 20th step be?<br>
#1 #2 #3 #4<br>

<table>
<tr><th>Design #</th><th># of Square</th></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>2</td></tr>
<tr><td>3</td><td>3</td></tr>
<tr><td>4</td><td>4</td></tr>
</table>
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Pattern and Relationship</td>
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<tr>
<td>B4.2.1.3.<br><br>Represent, describe and extend patterns and relationships, using charts and tables, to solve problems</td>
<td>
B4.2.1.3.1 Translate the information in a given problem into a table or char<br><br>
E.g. 1. Given the numerical pattern, ask students to extend the pattern and explain how they determined the pattern and its missing elements<br><br>
E.g. 2. Ask students to use manipulatives to represent this pattern and describe the relationship between the chart and the concrete representations<br><br>
<table>
<tr><td>A</td><td>B</td></tr>
<tr><td>1</td><td>2</td></tr>
<tr><td>2</td><td>4</td></tr>
<tr><td>3</td><td></td></tr>
<tr><td>4</td><td>8</td></tr>
<tr><td></td><td></td></tr>
</table>
B4.2.1.3.2 Identify and extend the patterns in a table or chart to solve a given problem.<br><br>
E.g. 1. Give students two tables with patterns that are related in some way. Students should first extend each one by three more numbers. Then they should explain how the two patterns are alike and how they are different<br><br>
E.g. 2. Students could make up their own pairs of patterns and challenge other students to discover how they are alike and how they are different<br><br>
<table>
<tr><td colspan="2">Pattern A</td></tr>
<tr><td>1</td><td>5</td></tr>
<tr><td>2</td><td>10</td></tr>
<tr><td>3</td><td>15</td></tr>
<tr><td>4</td><td>20</td></tr>
<tr><td>5</td><td>25</td></tr>
</table>
<table>
<tr><td colspan="2">Pattern B</td></tr>
<tr><td>1</td><td>2</td></tr>
<tr><td>2</td><td>7</td></tr>
<tr><td>3</td><td>12</td></tr>
<tr><td>4</td><td>17</td></tr>
<tr><td>5</td><td>22</td></tr>
</table>
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Pattern and Relationship</td>
</tr>
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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.2.2.1**<br>**Express a given problem as an equation in which a symbol is used to represent an unknown number** | **B4.2.2.1.1 Write a given problem as an equation in which a symbol is used to represent an unknown number**<br>E.g. 1. Ask students to explain the purpose of the box symbol in the following equation:<br>15 – □ = 8<br><br>**B4.2.2.1.2 Express a given pictorial or concrete representation of an equation in symbolic form**<br>E.g. 1. Provide a pictorial representation of the following and ask students to write equations with unknowns for each situation. Students need not solve their equations.<br>(i) The perimeter of a triangle is 12 cm. One side is 3 cm and another side is 4 cm. What is the length of the third side?<br>(ii) Mansa has three types of stickers. She has 25 stickers, 3 are Cocoa stickers and 18 are Cashew stickers. How many of the third type does she have?<br>(iii) Kojo's age and his sister's age add up to 18. If Kojo is 12, how old is his sister? | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Pattern and Relationship |



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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.2.2.2**<br>**Solve one-step equations involving a symbol to represent an unknown number** | **B4.2.2.2.1 Solve a given one-step equation using manipulatives**<br>E.g. 1. Ask students to solve the following using a pan balance:<br>(i) 7 + ☐ = 12 (ii) 19 – ☐ = 11<br>**B4.2.2.2.2 Describe orally, the meaning of a given one-step equation with one unknown.**<br>E.g. 1. Ask students to solve the following equations and explain their thinking.<br>(i) △ – 13 = 20 (ii) 25 + △ = 100<br>**B4.2.2.2.3 Solve a given equation when the unknown is on the left or right side of the equation.**<br>E.g. 1. Ask students to solve the following using base ten materials:<br>(i) 109 + ☐ = 164<br>(ii) △ – 50 = 150<br>**B4.2.2.2.4 Solve a given one-step equation using "guess and check"**<br>E.g. 1. Ask students to explain what the ☐ represents in the number sentences shown. Ask them to place numbers in the symbols to make the number sentence true.<br>△ - 7 = 6 9 + ☐ = 17<br>**B4.2.2.2.5 Identify the unknown in a problem, represent the problem with an equation, and solve the problem concretely, pictorially or symbolically**<br>E.g. 1. Provide students with linking cubes to model situations. Pose the problem: Kobby has 13 red marbles and 22 blue marbles. How many more blue marbles than red marbles does Kobby have? Observe how students solve the problem | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Pattern and Relationship |



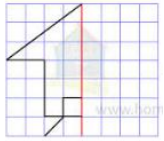
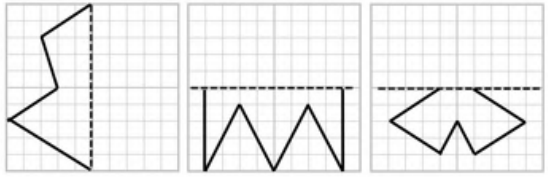
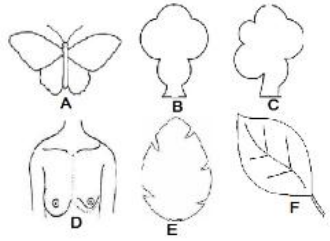
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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.2.2.2**<br>**Solve one-step equations involving a symbol to represent an unknown number. CONT'D** | **B4.2.2.2.6 Represent and solve a given addition or subtraction problem involving a "part-part-whole" or comparison context, using a symbol to represent the unknown**<br>E.g. 1. Ask students to represent and solve these problems:<br>(i) Eunice is Anita's older sister. The difference between their ages is 21 years. Anita is 37 How old is Eunice?<br>(ii) Ms. Akoto allowed 7 students to go to the washroom. There were 15 students left in the room. How many students are in the class?<br>**B4.2.2.2.7 Create a problem for a given equation with one unknown.**<br>E.g. 1. Provide an equation such as 14 + ☐ = 21 or 25 - ☐ = 18. Ask students to create a matching story problem and solve it. Observe to what extent students are able to:<br>-create a story to match the equation;<br>- explain the meaning of the unknown variable; and<br>- solve the problem in one or more way | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision, Pattern and Relationships |



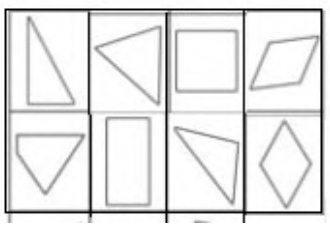
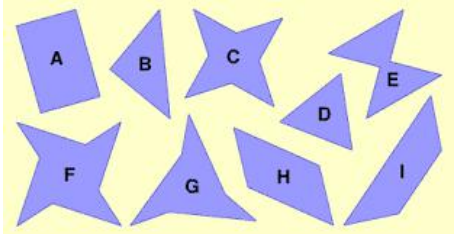
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**Strand 3: GEOMETRY AND MEASUREMENT**
**Sub-strand 1: 2D and 3D Shapes**

| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.1.1**<br><br>**Identify the lines of symmetry of regular and irregular 2D shapes** | **B4.3.1.1.1 Complete drawings of shapes to make them symmetrical**<br><br>E.g. 1. Give learners graph sheets to copy the incomplete 2D shape and it complete to make a symmetrical picture<br><br>E.g. 2. Give learners photocopied worksheets with several incomplete 2D shapes to complete with marked line of symmetry to complete the drawing.<br><br>  <br><br>  <br><br>**B4.3.1.1.2 Identify the lines of symmetry of regular and irregular 2D shapes (triangles and quadrilateral)**<br><br>E.g. 1. Give learners photocopied worksheets with symmetrical and non-symmetrical objects to sort and explain why they are symmetrical or not showing the lines of symmetry<br><br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



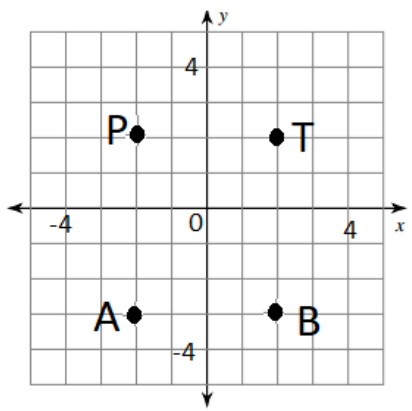
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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.1.1**<br>**Identify the lines of symmetry of regular and irregular 2D shapes** | E.g. 2. Give learners photocopied worksheets with symmetrical and non-symmetrical 2D shapes to cut out and sort, and explain why they are symmetrical or not showing the lines of symmetry<br>  <br>E.g. 3 Give learners cut-out 2D fold shapes to investigate the number of lines of symmetry<br><br>E.g. 4 Give learners 2-D shapes to copy and draw their lines of symmetry<br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



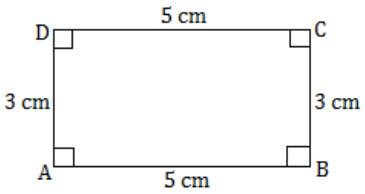
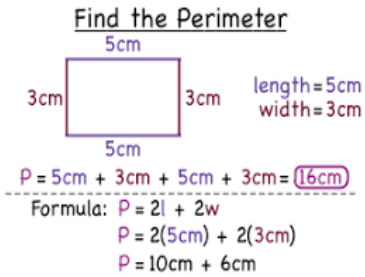
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**Sub-strand 2: Position / Transformation**

| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.2.1**<br>**Describe the position of objects in space using the cardinal points** | **B4.3.2.1.1 Tell the position and motion of objects in space using the cardinal points north, south, east and west**<br><br>E.g. 1. Put pupils in convenient groups a give each group with a worksheet on graph with different labelled coordinates; some with axes labelled and other not etc. Pupils discuss their worksheets in their groups, then with whole class, describing coordinates, axes and pointing out errors on graphs if any<br><br>E.g. 2. Pupils describe the position and motion of objects: e.g. The point **P** is to the west of point **T** but to the north of the **A**<br><br>  <br><br><br>E.g. 3. Ask learners to locate the coordinates (i.e. ordered pairs of the location) of the Points; e.g. B is (2,-3) and T is (2,2) | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

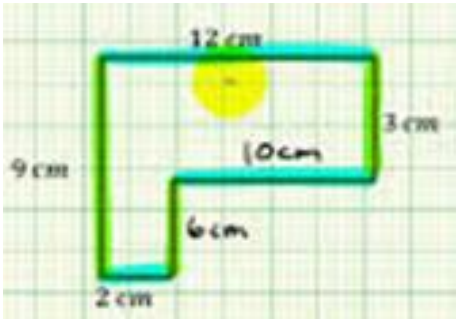
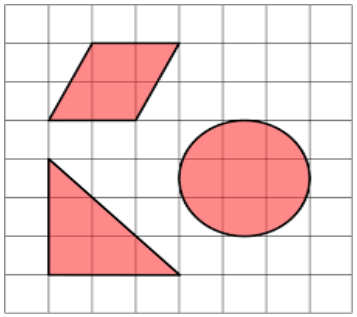
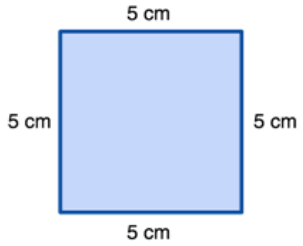
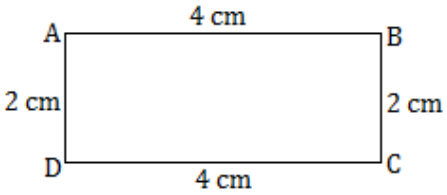


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.3.1**<br>**Demonstrate understanding of perimeter of regular and irregular shapes** | **B4.3.3.1.1 Estimate perimeter using referents for centimetre or metre**<br>E.g. 1 Using pupils' referents for centimetre or metre estimate and calculate perimeter of any object in the classroom (e.g. exercise book, floor tiles, math set, etc.) then measure with a ruler of tape and calculate the actual perimeter and compare the answers<br>  <br>**B4.3.3.1.2 Measure and record perimeter for regular and irregular shapes in cm and m.**<br>E.g. 1 Ask pupils to use a ruler to measure all the sides of a regular shape and sum up the result to obtain the perimeter<br>  <br><br>P = 16cm<br>  <br><br>E.g. 2 Pupils to explore the area of irregular cut-out shapes using graph sheets. Pupils to trace the shape on the graph sheet and count out the unit squares | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; Cultural Identity |

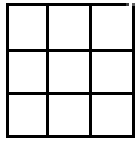
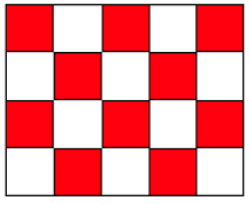


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.3.1 Demonstrate understanding of perimeter of regular and irregular shapes CONT'D** | E.g. 3 Given an irregular shape with its sizes, ask pupils to sum up the sizes to obtain the perimeter.<br>  <br>  <br>**B4.3.3.1.3 Develop and apply a formula for determining perimeter of square and rectangle**<br>E.g. 1. Use cut out shapes to demonstrate how to determine the perimeter of 2-D shapes. Ask pupils to measure the sides of the cut out shape and add the results to obtain the perimeter<br>  <br>E.g. 2 Given a rectangular cut out shape, ask pupils to measure the sides of the shape and determine the perimeter of the shape<br>  <br>Therefore the perimeter of the triangle given is<br>AB + AD + BC + CD. Thus 4cm + 2cm + 4cm + 2cm = 12cm | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision: Patterns and Relationships |

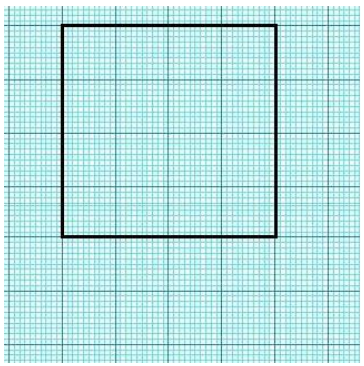


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B4.3.3.1 Demonstrate understanding of perimeter of regular and irregular shapes CONT'D** | **B4.3.3.1.4 Construct different rectangles for a given perimeter (cm, m) to demonstrate that many shapes are possible for a perimeter.**<br>E.g. 1. Given a perimeter of say 36cm, pupils to draw about three different rectangular shapes for the same perimeter. Combinations may include (6cm, 6cm, 12cm, 12cm; 8cm, 8cm, 10cm, 10cm; 4cm, 4cm, 14cm, 14cm, etc.) | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Personal Development and Leadership Attention to Precision |
| **B4.3.3.2**<br><br>**Demonstrate an understanding of area of regular and irregular 2-D shapes** | **B4.3.3.2.1 Recognise that area is measured in square units.**<br>E.g. 1 Pupils to explore the area of their exercise books. i.e. ask pupils to cut 1cm by 1cm square sheets and tack them on the back of their exercise books until it is covered entirely. Then, count the number of the sheets to obtain the area<br>E.g. 2. Pupils to explore the area of graph sheet. Ask pupils to calculate the area of one of the squares on the graph sheet. Then, count the number of squares on the graph sheet to obtain the area of the entire sheet<br>  <br>Area of graph sheet = $9cm^2$<br>E.g. 3 Using floor tiles, ask pupils to measure the area of one of the tiles, then multiply it by the number of tiles to obtain the area of the entire room<br>  <br>Note: Help pupils to come out with the understanding that the area of a figure is the number of squares required to cover it completely<br><br>**B4.3.3.2.2 Select and justify referents for the units $cm^2$ or $m^2$**<br>E.g. 1 Let pupils explore the area of smaller objects say their exercise books. Then explore the area of larger objects say the entire classroom and compare the results for the two activities.<br><br>Ask pupils to tell which unit ($cm^2$ or $m^2$) they will use for which activity and why? | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision |

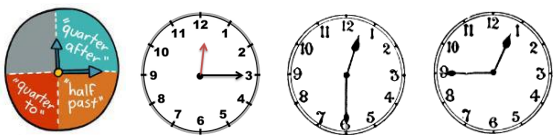
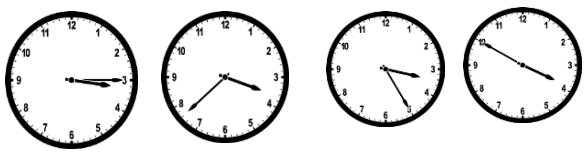


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|---|---|---|
| **B4.3.3.2**<br><br>**Demonstrate understanding of area of regular and irregular 2-D shapes CONT'D** | **B4.3.3.2.3 Estimate area by using referents for $cm^2$ or $m^2$**<br><br>E.g. 1 Let pupils estimate the area of items in the classroom. Then measure the actual area and compare with their estimates<br><br>E.g. 2 Pupils to measure the area of say their exercise books. With that as a reference point, ask them to estimate the area of the teacher's table<br><br>**B4.3.3.2.4 Develop and apply a formula for determining area of a rectangle and square**<br><br>E.g. 1 Give pupils a standard graph sheet with 1 cm squares and ask them to draw a square of given length on the graph sheet. The area can be measured by counting the number of square units. So, we can count the number of unit units to find the area. In the diagram the area is $16cm^2$ since there are 36 squares in the shape<br><br>  <br><br>E.g. 2 Ask pupils to count the number of cubic squares on the vertical section of the shape and multiply it by the number on the horizontal section<br><br>Ask pupils to compare their answer with the first strategy and draw their own conclusions.<br>The area of a square can be given as L × L and that of a rectangle is L × B.<br><br>**B4.3.3.2.5 Construct different rectangles for a given area ($cm^2$ or $m^2$) in order to demonstrate that many different rectangles may have the same area.**<br><br>E.g. 1. Given an area of say $24cm^2$, pupils should draw about three different rectangles for the same area. Combinations may include (6cm × 4cm; 8cm × 3cm; 12cm × 2cm, etc.) | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| **B4.3.3.3 Demonstrate understanding of time taken by events in minutes and hours** | **B4.3.3.3.1 Tell the time in hours and minutes in analogue and digital watches including 24-hour clocks**<br><br>E.g .1 Use analogue clocks to tell time by keeping the hour hand on 12 and making jumps of 15(quarter past), 30(half past), 45(quarter to), and 60 (12 o'clock)<br>  <br>E.g. 2 Use analogue clocks to tell time by keeping the hour hand on any number say 3 and move the minutes hand to various numbers for different time (3:12, 3:25, 3:38, 3:50, etc.<br>  <br>E.g. 3 Use analogue clocks to tell time by keeping the minutes hand on 12 and moving the hour hand to tell time at the hour mark<br>  <br>E.g .4 Pupils should draw/sketch the face of the clock to tell time given | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision Cultural Identity and Global Citizenship |



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| **B4.3.3.3 Demonstrate an understanding of time taken by events in minutes and hours CONT'D** | E.g. 5 Pupils to explore the relationship in the units of time (i.e. 60 seconds = 1 minute; 60 minutes = 1hour)<br><br>E.g. 6 Use digital watches to tell time. Pupils to observe the face of the digital watches and tell the time<br><br>  <br><br><br>**B4.3.3.3.2 Use clock to measure time to complete simple events in minutes and seconds**<br><br>E.g. 1 Use the analogue clock to record the time it takes to say walk from the classroom to the canteen/head teacher's office/library, etc. and back. (mostly minutes)<br><br>E.g. 2 Use the digital watch to record say the time it takes to walk from the teacher's table to the cupboard. (mostly seconds)<br><br>E.g. 3 Ask pupils to tell the time it takes to complete activities like bathing, brushing teeth, ironing a shirt, cooking rice, etc.<br><br>**B4.3.3.3.3 State dates of events and record calendar dates in a variety of formats**<br>E.g. 1. Record dates in different formats<br><br>i. Tuesday, 28th August 2018<br>ii. 28th August, 2018<br>iii. 28-Aug-18<br>iv. 28/08/2018<br>v. 28.08.18<br><br>E.g. 2. Ask pupils to write the dates for the following events from a calendar;<br><br>i. Independence Day of Ghana<br>ii. Republic Day<br>iii. Founders' Day<br>iv. Farmers' Day<br>v. Workers' Day | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision Cultural Identity and Global Citizenship |

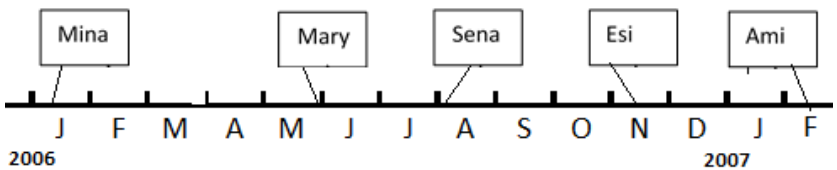


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| CONTENT STANDARDS | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B4.3.3.3 Demonstrate an understanding of time taken by events in minutes and hours CONT'D** | E.g. 3. Ask pupils in groups to state their dates of birth and put this in a table. Guide pupils to draw a time line (or time line-graph) for their birthdays (see Figure below) | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision Cultural Identity and Global Citizenship |

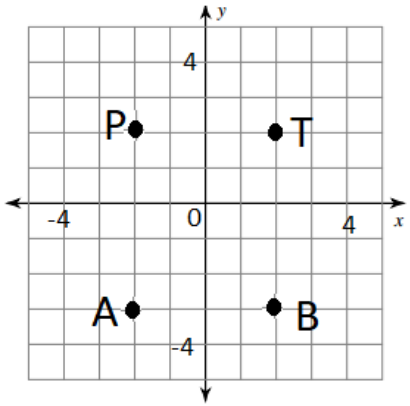
Table 1 Birthdays of pupils in Ami's group

| Name | Date of birth |
|---|---|
| Ami | 20th February 2007 |
| Esi | 14th November 2006 |
| Mary | 30th May 2006 |
| Mina | 10th January 2006 |
| Sena | 5th August 2006 |



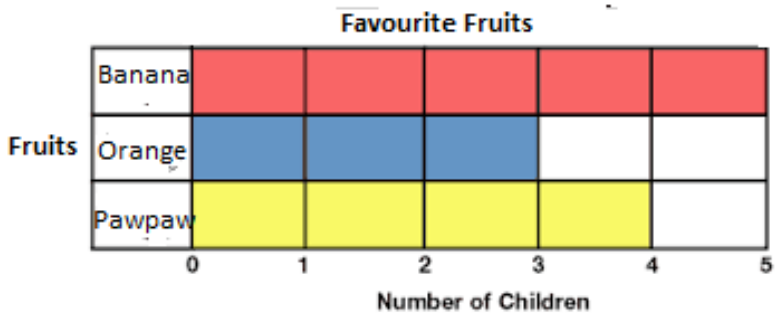



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|---|---|---|
| **B4.3.3.1**<br><br>**Describe the position of objects in space using the cardinal points** | **B4.3.3.1.1**<br><br>**Tell the position and motion of objects in space using the cardinal points north, south, east and west**<br><br>E.g. 1. Put pupils in convenient groups a give each group with a worksheet on graph with different labelled coordinates; some with axes labelled and other not etc. Pupils discuss their worksheets in their groups, then with whole class, describing coordinates, axes and pointing out errors on graphs if any<br><br>E.g. 2. Pupils describe the position and motion of objects: e.g. The point **P** is to the west of point **T** but to the north of the **A**<br><br>E.g. 3. Ask learners to locate the coordinates (i.e. ordered pairs of the location) of the Points; e.g. B is (2,-3) and T is (2,2)<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Personal Development and Leadership Attention to Precision, Global Citizenship |



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| **B4.4.1.1**<br><br>**Demonstrate an understanding of many-to-one correspondence in displaying, and reading or interpreting, graphs** | **B4.4.1.1.1 Use an understanding of one-to-one correspondence to read and interpret graphs**<br><br>E.g. 1. Give learners graphs that use one-to-one correspondence in displaying data. For example, the graph below shows the favourite fruits of children in a P4 class. Each coloured box in the graph represents a pupil<br><br>  <br><br><br>Ask the following questions for children to read and interpret graphs<br><br>i. How many pupils said they like pawpaw?<br>ii. What is the most favourite food of the class?<br><br>How many pupils are in the class? Etc. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision; |

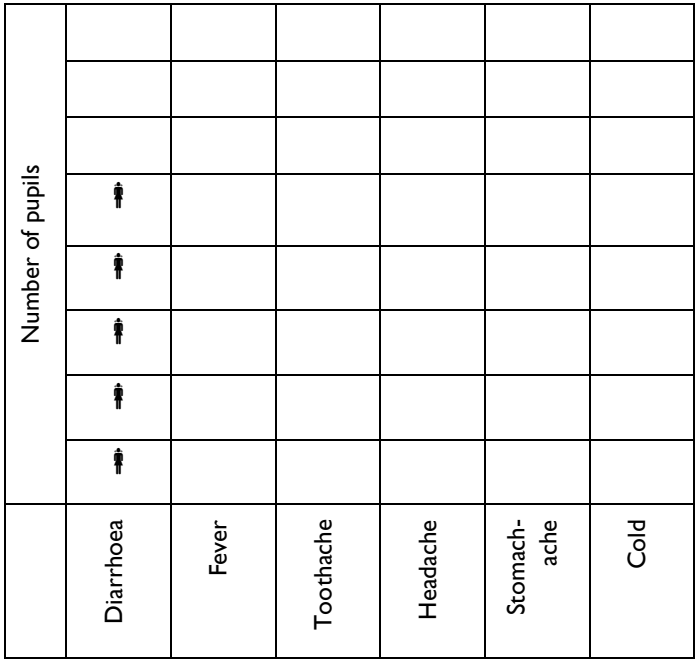


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<tr><td>B4.4.1.1<br>Demonstrate an understanding of many-to-one correspondence in displaying, and reading or interpreting, graphs CONT'D</td><td>B4.4.1.1.2 Use an understanding of many-to-one correspondence to display or construct graphs<br>E.g. 1. Display a table for of illnesses P4 pupils have had in school last academic year. Discuss with pupils how many objects/cut-outs will be required to display or construct a graph of the illnesses in the table if 🧍 represents 2 pupils<br><table><tr><th>Illness</th><th>Number of pupils visiting hospital with illnesses</th><th>Number of objects 🧍 in graph</th></tr><tr><td>Diarrhoea</td><td>10</td><td>5</td></tr><tr><td>Fever</td><td>16</td><td>8</td></tr><tr><td>Toothache</td><td>4</td><td>2</td></tr><tr><td>Headache</td><td>6</td><td>·</td></tr><tr><td>Stomach-ache</td><td>8</td><td>·</td></tr><tr><td>Cold</td><td>14</td><td>·</td></tr></table><br>E.g. 2. Put a chart with labelled axes on the board; places several card cut-outs of the object " 🧍 "on the table. Ask pupils to go in turns to construct the graph of the illnesses in the chart by pasting the right number of cut-outs in the columns above each illness as in the figure below</td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision</td></tr>
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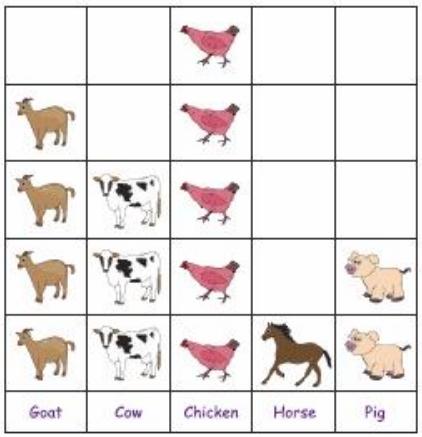
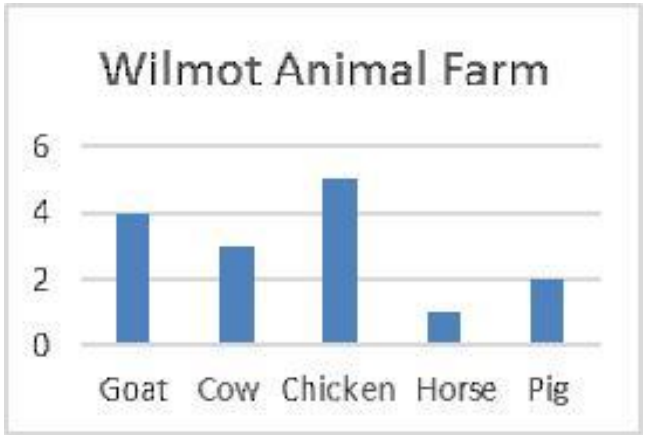


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|---|---|---|
| **B4.4.1.1**<br>**Demonstrate an understanding of many-to-one correspondence in displaying, and reading or interpreting, graphs CONT'D** |  <br>E.g. 3. Ask questions for pupils to read and interpret graphs. Ask pupils also to write questions based on the graph for their friends to answer<br><br>E.g. 4. Give learners tables with data that require the use an understanding of many-to-one correspondence to construct their graphs | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Attention to Precision |

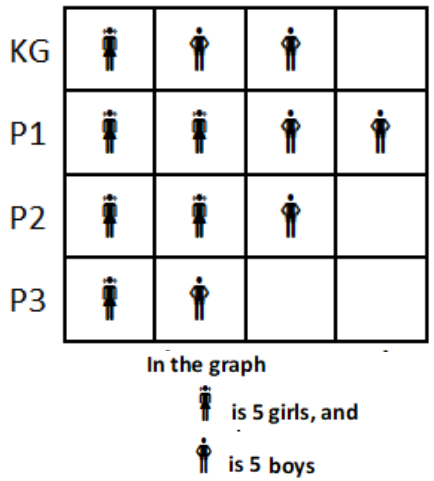
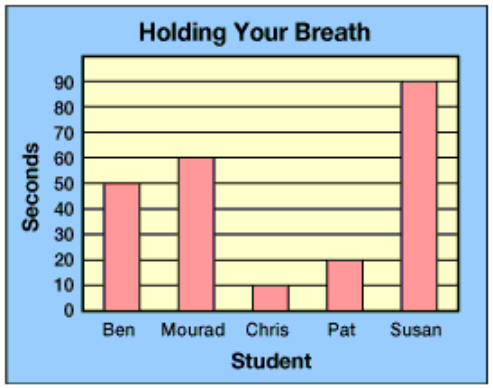


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|---|---|---|
| **B4.4.1.1 Demonstrate an understanding of many-to-one correspondence in displaying, and reading or interpreting, graphs CONT'D** | **B4.4.1.1.3 Compare graphs in which the same data has been displayed and explain how they are the same and different**<br>E.g. 1. Give learners several graphs that have used one-to-one correspondence in displaying the same data. For instance, the graph below shows the animals at Mr. Wilmot's farm. Ask children to explain how they are the same and different<br>  <br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Attention to Precision |

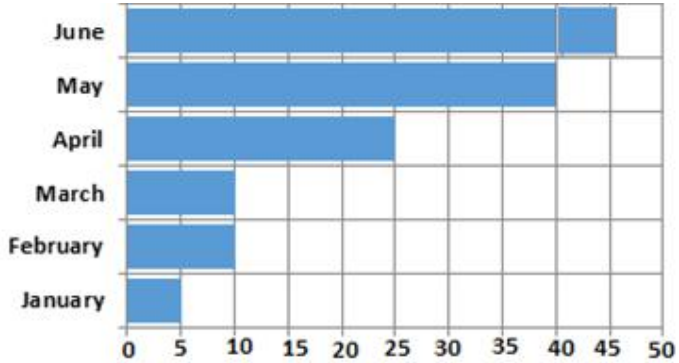


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| **B4.4.1.1 Demonstrate an understanding of many-to-one correspondence in displaying, and reading or interpreting, graphs CONT'D** | **B4.4.1.1.4 Find examples of graphs in which many-to-one correspondence is used in print and electronic media, such as newspapers, magazines and the Internet, and describe the correspondence used**<br>E.g. 1. Give learners several graphs that have used many-to-one correspondence in displaying the same data. E.g. The graphs below show the number of pupils in a KG to P3 in a school and the time five pupils can take to hold their breath. Ask questions for children to read and interpret graphs<br>  <br>  <br>How many pupils are in KG and P1?<br>How many pupils more pupils are in P1 than P3?<br>How long did Ben hold his breath?<br>Who can hold her breath longest? Etc. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Attention to Precision |



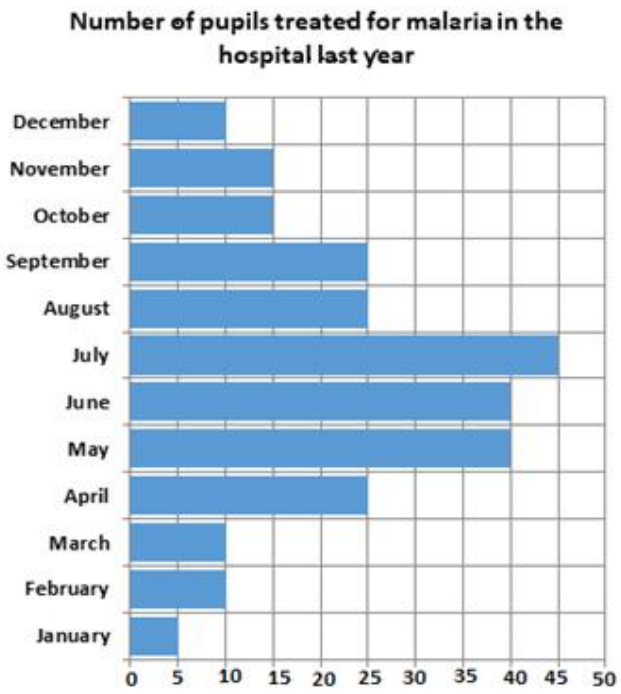
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<tr>
<td>B4.4.1.2:<br><br>Construct and interpret pictographs and bar graphs involving many-to-one correspondence to draw conclusions.</td>
<td>
B4.4.1.2.1 Identify common features of graphs that use many-to-one correspondence and use that understanding to draw bar graphs or pictographs, complete with title, labelled axes, key or legend, to represent data collected (up to 4 categories of data).<br><br>
E.g. 1. Refer to example in B4.4.1.1.4 to common features of graphs that use many-to-one correspondence. Examine such features as title, vertical axis and label, hofrizontal axis and label, key or legend.<br><br>
E.g. 2. Give students data presented in a table to draw bar graph complete with title, labelled axes, key. The table show amount of rainfall recorded in Kumasi in the half of the year.<br><br>
<table>
<tr><td>January</td><td>February</td><td>March</td><td>April</td><td>May</td><td>June</td></tr>
<tr><td>5mm</td><td>10mm</td><td>10mm</td><td>25mm</td><td>40mm</td><td>45mm</td></tr>
</table>


<br>
E.g. 3. Give pupils data on energy usage by households in a community over a period (e.g. six months) presented in tables for them to draw bar graph complete with title, labelled axes, key. Ask questions for pupils to read and interpret graphs.
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Attention to Precision</td>
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| **B4.4.1.2:**<br>**Construct and interpret pictographs and bar graphs involving many-to-one correspondence to draw conclusions CONT'D** | **B4.4.1.2.2 Use understanding of many-to-one correspondence to solve simple problems (how many altogether, comparing, or take apart problems) requiring interpretation of many-to-one bar graphs (up to 4 categories of data).**<br>E.g. 1. Give learners several graphs that have used many-to-one correspondence in displaying the same data. For example, the graph below on the number of pupils treated for malaria in a school. Ask questions for children to read and interpret graphs.<br>  <br><br>i) How many pupils were treated for malaria in the hospital in April?<br>ii) How many pupils were treated for malaria in the hospital in May and June?<br>iii) How many pupils were treated for malaria in the hospital in the last three months of the year?<br>iv) In which month was the largest number of people treated for malaria in the hospital? | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Attention to Precision |



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